Daily Drilling Report
SPD19B-03
4/9/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
15.0	16"	36.00	72.83	31.70	98.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
18 5/8" Shoe, 870.00mKB	204,530.15	343.72	3,530,489.90		12.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
878.00	960.00	926.61	82.00	6.70	12.2

Operations at Report Time
Drilling @ 1012m.

Operations Summary
Make enough 8 1/4" DC and 5 7/8" DP. RIH 16" Directional BHA#5. Drilling 16" hole section f/ 878m to 960m.

Operations next 24hrs
Cont' drilling 16" hole.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Vahab Kari	Night Drilling Sup.
Behnam Jahangiri	Drilling Engineer
Ali Sahebnouri	Geologist
M. H. Hemmati	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	3/9/2014	4/9/2014
BOP Test	4/7/2014	4/29/2014
Chemical Spill Drill	3/29/2014	5/29/2014
Fire Drill	3/9/2014	4/9/2014
H2S Drill	4/4/2014	5/5/2014
Heli Crash Drill	3/22/2014	5/22/2014
Medical Drill	4/4/2014	6/4/2014
MOB Drill	2/28/2014	4/30/2014
Pit Drill	4/9/2014	5/25/2014
Trip Drill	4/8/2014	5/29/2014

STOP Cards

Number of Reports: 34

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	08:00	8.00	16"	D	01	P	RIH 1 jt 9 1/2" NMDC + 9 1/2" Float sub + 8 jts 8 1/4" DC + 18 jts 5 7/8" DP and flush. POOH and R/B on derrick.
08:00	10:00	2.00	16"	D	04	P	P/U and M/U 16" Directional BHA#5 consists of below items, RIH to 24m. Align motor with MWD. * 16" PDC bit (VM516DGXU, M322, SN: 6004481, Nozzles: 8 x 13) * 9 5/8" Mud Motor (BH: 1.25 , Sleeve Stabilizer: 15 3/4") * 14 3/4" String Stabilizer (COR3032AD) * 9 1/2" Restrictor sub (Jet = 36) * 9 1/2" NMDC * 9 1/2" TM - Sub (Put MWD and gamma probes inside)
10:00	10:30	0.50	16"	D	11	P	Connect TDS. Perform MWD and Motor Shallow Test w/ 700 gpm / 720 psi , OK. * Perform shallow test with tricone bit.
10:30	13:00	2.50	16"	D	04	P	Cont' RIH rest of BHA items as per below to 313m. * 1 jt 9 1/2" NMDC * 9 1/2" Float sub with valve inside * XOS * 8 jts 8 1/4" DC * 8" DA Cougar Jar (COR2898AD) * 6 jts 8 1/4" DC * XOS * 15 jts 5 7/8" HWDP
13:00	15:00	2.00	16"	D	04	P	RIH 16" directional BHA on 5 7/8" DP f/ 313m to 865m. W/D to 878m and tag bottom.
15:00	00:00	9.00	16"	D	03	P	Profile new bit. Directional drilling 16" hole in rotation mode f/ 878m to 960m w/ 750-900 gpm/ 1650-2100 psi/ 5-22 kips/ 50-60 rpm/ 8 - 17 kft-lbf. ** Drilling with NL / NG. ** B/R and R/D each stand drilled once. ** Pump 20 bbl Hi-Vis pill when stand down. ** Take survey before connection. >>> Note: One of rig centrifuges is out of service.

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Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:00						Apr 10 , 2014 00:00 - 06:00 Cont' directional drilling 16" hole in rotation and slide mode f/ 960m to 1012m w/ 850 gpm/ 2000 psi/ 5-18 kips/ 60 rpm/ 10 - 16 kft-lbf. Rotation Intervals Slide Interval 960m - 984m 984m - 988m 988m - 1012m ** Drilling with NL / NG. ** B/R and R/D each stand drilled once. ** Pump 20 bbl Hi-Vis pill when stand down. ** Take survey before connection. >>> Last sample @ 1000m consists of : Limestone (100 %)

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
960.00	Polymer Mud	8.70	50	16.0	15.000	6.0
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
9.8						
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
3	5	0.0	95.0	5.0	23,030.000	
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
3.000	4.000		0/100	38.0	1236.0	1427.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
620.0	52.0	0.0			6.654657	0.511

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.20	7.	70	1.0	1,008.0	24.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
DP3	Supply Boat	4/9/2014 13:15	4/10/2014 02:40	Arrived DD6 from Kish, Delivered all DD6 Cargo, Dep for Aban 8.
Marina	Supply Boat	4/9/2014 06:50	4/9/2014 11:20	Arrived DD6 from Aban 8. Delivered Cement-30 MT, Dep for Sahar 2.
Ahmed-06	Supply Boat	4/9/2014 21:30		Arrived DD6 from Sahar 1 at 2130 LT, Stdby Anchorage.
Morshed	Supply Boat	4/8/2014 21:40	4/9/2014 18:30	Delivered bulk (FO, CMT, DW, Bentonite), B/L cargo. departed to Kish.
Swissco	Supply Boat	4/9/2014 20:10	4/10/2014 01:40	Arrived DD6 from Kharg Island. Del Cmt-60 MT & Barite-59 MT, Dep for Sahar 1.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Bentonite Bulk		216.0	73.0
Bio-XCD	1.0	414.0	162.0
Defoam SLC	2.0	220.0	77.0
Caustic Soda	3.0	657.0	152.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Barite BB	0.0	0.0	0.0
Barite Bulk	0.0	59.0	147.0
Cement G	0.0	118.0	150.0
Bentonite Bulk	0.0	40.0	65.0
Bentonite BB	0.0	0.0	21.0
Diesel Fuel	12.0	70.0	163.0
Drill Water	16.0	300.0	700.0
Potable Water	20.0	20.0	110.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
5	16	Varel	VM516DGXU	6004481	PDC	M322	13/13/13/13/13/13/13/13
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
1.04	6.70	66,531.60	878.00	960.00	82.00	12.2	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
750	890	1,650.0	2,100.0	10	22	148	176
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
9.28	6.70	100.00	0.00	290	267	237	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
878.00	892.00	1.43	9,000.0	750	50	1,650.0	10	267
892.00	922.00	2.02	11,000.0	850	50	1,980.0	22	267
922.00	960.00	3.25	14,000.0	890	60	2,100.0	18	267

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
15 jts 5 7/8" HWDP	5 7/8	4.250	139.60	313.30
XOS	8	3.000	0.79	173.70

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Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
6 jts 8 1/4" DC	8 1/4	2.810	56.24	172.91
8" DA Drilling Jar (COR2898AD)	8	3.500	5.72	116.67
8 jts 8 1/4" DC	8 1/4	2.810	75.34	110.95
XOS	9 1/2	3.000	1.22	35.61
9 1/2" Float sub with valve	9 1/2	3.000	0.78	34.39
9 1/2" NMDC	9 1/2	4.000	9.44	33.61
9 1/2" TM-Sub (Mwd and gamma probes inside)	9 1/2		2.11	24.17
9 1/2" NMDC	9 1/2	4.000	9.44	22.06
9 1/2" Restrictor Sub (Jet: 36)	9 1/2	4.300	0.92	12.62
14 3/4" String Stabilizer (COR3032AD)	14 3/4	3.000	2.16	11.70
9 5/8" Motor (BH: 1.25 , Sleeve: 15 3/4")	9 5/8		9.14	9.54
16" PDC Bit (SN: 6004481)	16		0.40	0.40

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.24	400.0

Applied Surface Pressure (psi)	Volume Pumped (bbl)	Current MAASP	Kick Tolerance	Dens Fluid (lb/gal)	Leak Off Pressure (psi)
400.0	8.0	400.00		11.48	1,650.3

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
873.28	249.50	21.60	846.30	-90.18	-152.65	177.23	0.18	0.18
902.25	247.60	21.90	873.21	-94.11	-162.64	187.87	0.79	0.31
931.34	247.20	22.40	900.16	-98.33	-172.76	198.77	0.54	0.52
959.10	247.10	22.80	925.78	-102.47	-182.60	209.38	0.43	0.43
989.20	247.10	22.10	953.60	-106.94	-193.18	220.81	0.70	-0.70

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Asmari		374.50	372.44
Jahrum		493.00	487.29
Ilam		846.00	820.89
Laffan		939.00	907.23
Sarvak		972.00	937.70

General Remarks

Comment

>>> Requirements: (50 MT Cement)

>>> Sent:

** Morshed to Kish: (2 x Empty food container)

>>> Received:

** Swissco f/ Kish: (59 MT Barite) + (60 MT Cement)

** Morshed f/ Kish: (300 MT DW) + (28 MT Cement) + (40 MT Bentonite) + (70 MT FO)

** Marina f/ Kish: (30 MT Cement)

** DP3 f/ Kish: (154 jts 13 3/8" CSG) + (3 no 13 3/8" Float shoe) + (3 no 13 3/8" Float collar) + (130 x Bow centralizer) + (260 x Stop collar) + (3 x Top cement plug) + (3 x Bottom cement plug) + (6 x Rigid centralizer) + (8 x Thread compound) + (5 no 13 3/8" Coupling) + (2 no Thread lock) + (23 no 13 3/8" Pup Joint) + (2 no 16" insert bit) + (2 no Split Unihead assembly) + (2 no Casing hanger R/R tool) + (1 no Drift mandrel for 68 ppf 13 3/8" casing).